Attaching to Docker Containers

To Begin with the Lab

Summary of the Lab

This lab demonstrated Docker container lifecycle management by running NGINX and Ubuntu containers in different modes (detached, interactive, and attached). It covered listing running and stopped containers, attaching to containers, stopping containers, executing Linux commands inside a container, and safely detaching without stopping it, highlighting the importance of long-running processes in keeping containers active.

- Run an NGINX Container
- Start an NGINX container in detached mode with port mapping.

```
sudo docker run -d -p 80:80 nginx
```

- Docker returns the Container ID.
- List Running Containers

```
sudo docker ps
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d -p 80:80 nginx
2750e4b5b98f6a43616e2441a4b15419a219f8108687ea49f22fa8514ca5e788
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                 NAMES
2750e4b5b98f   nginx     "/docker-entrypoint.…"   21 minutes ago   Up 21 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp   practical_greider
ubuntu@ip-172-31-45-84:~$
```

- Note the Container ID of the running NGINX container.
- Attach to the NGINX Container
- Attach to the running container using the first few characters of the Container ID.

```
sudo docker attach 275
```

- You are attached to the container.
- Since NGINX is the running process, you cannot interact with the container.

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d -p 80:80 nginx
f7552a7519a21a33e1931cedfcdcba4b0a5af6ce2ebd683987f42440d5081448
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                 NAMES
f7552a7519a2   nginx     "/docker-entrypoint.…"   4 seconds ago   Up 4 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   zealous_khayyam
ubuntu@ip-172-31-45-84:~$ sudo docker attach f755
```

- Any output generated by the container will be displayed.
- Open a New Session
- Verify the Running Container

```
sudo docker ps
```

- The NGINX container should still be running.

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jul  8 04:19:17 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   15.2% of 18.25GB   Processes:             132
  Memory usage: 35%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.45.84

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

58 updates can be applied immediately.
47 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Wed Jul  8 03:49:46 2026 from 18.206.107.29
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                 NAMES
f7552a7519a2   nginx     "/docker-entrypoint.…"   2 minutes ago   Up 2 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp   zealous_khayyam
ubuntu@ip-172-31-45-84:~$
```

- Stop the NGINX Container

```
sudo docker stop f75
```

- Run an Ubuntu Container
- Run a container using the Ubuntu image.

```
sudo docker run ubuntu
```

- If the image is not available locally, Docker automatically downloads it from Docker Hub.

```
ubuntu@ip-172-31-45-84:~$ sudo docker run ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
2c1ce1d0a589: Pull complete
a9be9fd915e9: Pull complete
bce5ee84b2b4: Download complete
Digest: sha256:b7f48194d4d8b763a478a621cdc81c27be222ba2206ca3ca6bc42b49685f3d9e
Status: Downloaded newer image for ubuntu:latest
ubuntu@ip-172-31-45-84:~$
```

- Check the Container Status

```
sudo docker ps
```

- No running containers should be displayed.
- View all containers:

```
sudo docker ps -a
```

- The Ubuntu container will have the status Exited.

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-45-84:~$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS                      PORTS     NAMES
f580899d95b6   ubuntu    "/bin/bash"              4 minutes ago    Exited (0) 4 minutes ago              gifted_brahmagupta
9791cea36102   ubuntu    "/bin/bash"              5 minutes ago    Exited (0) 5 minutes ago              youthful_sinoussi
f7552a7519a2   nginx     "/docker-entrypoint.…"   10 minutes ago   Exited (0) 6 minutes ago              zealous_khayyam
2750e4b5b98f   nginx     "/docker-entrypoint.…"   37 minutes ago   Exited (0) 14 minutes ago             practical_greider
7c179fd50eb7   nginx     "/docker-entrypoint.…"   18 hours ago     Exited (137) 17 hours ago             wonderful_wiles
ubuntu@ip-172-31-45-84:~$
```

- Run Ubuntu in Detached Mode

```
sudo docker run -d ubuntu
```

- Check the container status:

```
sudo docker ps
```

- Again, no running containers will be displayed.
- Verify with:

```
sudo docker ps -a
```

- The container has exited because there is no long-running process inside the Ubuntu image.

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d ubuntu
42b8a28dc825b08b8bc8405e80752c62f71f22103bdad6a2c58d18a98c9f7689
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-45-84:~$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS                       PORTS     NAMES
42b8a28dc825   ubuntu    "/bin/bash"              27 seconds ago   Exited (0) 27 seconds ago              pedantic_greider
f580899d95b6   ubuntu    "/bin/bash"              7 minutes ago    Exited (0) 7 minutes ago               gifted_brahmagupta
9791cea36102   ubuntu    "/bin/bash"              8 minutes ago    Exited (0) 8 minutes ago               youthful_sinoussi
f7552a7519a2   nginx     "/docker-entrypoint.…"   13 minutes ago   Exited (0) 9 minutes ago               zealous_khayyam
2750e4b5b98f   nginx     "/docker-entrypoint.…"   40 minutes ago   Exited (0) 17 minutes ago              practical_greider
7c179fd50eb7   nginx     "/docker-entrypoint.…"   18 hours ago     Exited (137) 17 hours ago              wonderful_wiles
ubuntu@ip-172-31-45-84:~$
```

- Run Ubuntu in Interactive Mode
- Run the Ubuntu container with the -it option.

```
sudo docker run -d -it ubuntu
```

- Verify the Running Container

```
sudo docker ps
```

- The Ubuntu container should now be in the running state.
- Attach to the Ubuntu Container

```
sudo docker attach 8a6
```

- You are now attached to the Ubuntu container as the root user.

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d -it ubuntu
8a6b8279ffdfdc9407e698b57010135e10ecda8ce574fdbb481be82d5662294a
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS     NAMES
8a6b8279ffdf   ubuntu    "/bin/bash"   8 seconds ago   Up 7 seconds             hopeful_dirac
ubuntu@ip-172-31-45-84:~$ sudo docker attach 8a6
root@8a6b8279ffdf:/#
```

- Execute Linux Commands
- Run a few basic Linux commands inside the container.
- List files:

```
ls
```

- View running processes:

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d -it ubuntu
8a6b8279ffdfdc9407e698b57010135e10ecda8ce574fdbb481be82d5662294a
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS         PORTS     NAMES
8a6b8279ffdf   ubuntu    "/bin/bash"   8 seconds ago    Up 7 seconds             hopeful_dirac
ubuntu@ip-172-31-45-84:~$ sudo docker attach 8a6
root@8a6b8279ffdf:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@8a6b8279ffdf:/#
```

```
top
```

- Detach from the Container

```
top - 04:34:56 up 45 min,  0 users,  load average: 0.00, 0.02, 0.00
Tasks:   2 total,   1 running,   1 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    908.7 total,    116.2 free,    384.2 used,    525.6 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    524.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0    4776   3976   3376 S   0.0   0.4   0:00.03 bash
     10 root      20   0    7752   5392   3336 R   0.0   0.6   0:00.00 top
```

- Without stopping the container, press:
    - Ctrl + P
    - Ctrl + Q
- This detaches from the container and returns you to the EC2 terminal.

```
top - 04:35:26 up 45 min,  0 users,  load average: 0.00, 0.01, 0.00
Tasks:   2 total,   1 running,   1 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.8 id,  0.2 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    908.7 total,    116.2 free,    384.2 used,    525.7 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    524.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0    4776   3976   3376 S   0.0   0.4   0:00.03 bash
     10 root      20   0    7752   5392   3336 R   0.0   0.6   0:00.00 top
read escape sequence
ubuntu@ip-172-31-45-84:~$
```

- Verify the Container Is Still Running

```
sudo docker ps
```

- The Ubuntu container should still be running.

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS     NAMES
8a6b8279ffdf   ubuntu    "/bin/bash"   4 minutes ago   Up 4 minutes             hopeful_dirac
ubuntu@ip-172-31-45-84:~$
```

Hosting a Simple HTML Web Application on Apache

To Begin with the Lab

Summary of the Lab

This lab demonstrated hosting a simple HTML website using the Apache web server on Ubuntu. It covered updating packages, installing and verifying Apache, accessing the web root directory, modifying permissions, replacing the default webpage with a custom index.html, verifying the changes in a browser, and stopping the Apache service after completion.

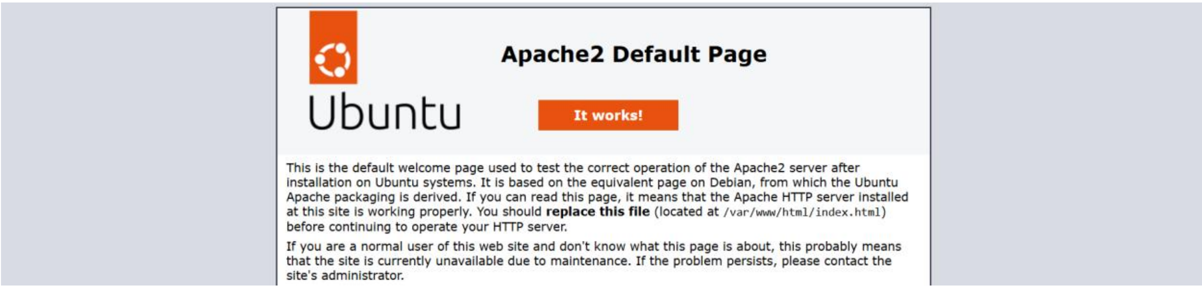
- Update the Package Index

```
sudo apt update
```

- Install Apache

```
sudo apt install apache2 -y
```

- Verify Apache
- Open your browser:
- http://<EC2-Public-IP>
- You should see the default Apache2 Ubuntu Default Page.



- Navigate to Apache's Web Root

```
cd /var/www/html
```

- List the files:

```
ls
```

- You should see:
- index.html

- Change Directory Permissions

- Give write permissions to the web root directory.

```
sudo chmod -R 777 /var/www/html
```

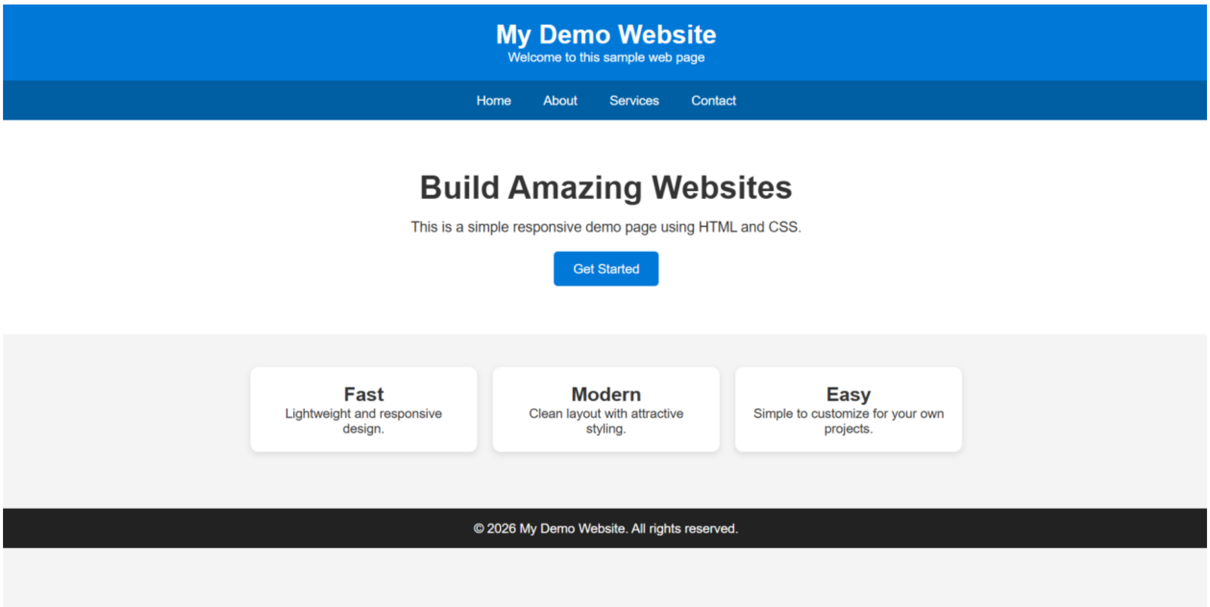
- Remove the Default Web Page
- sudo rm /var/www/html/index.html
- Verify:
- ls /var/www/html
- The default index.html should no longer exist.

```
ubuntu@ip-172-31-45-84:/var/www/html$ sudo chmod -R 777 /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ sudo rm /var/www/html/index.html
ubuntu@ip-172-31-45-84:/var/www/html$ ls /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ 
```

- Now create a new index.html in the same directory /var/www/html

```
ubuntu@ip-172-31-45-84:/var/www/html$ sudo chmod -R 777 /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ sudo rm /var/www/html/index.html
ubuntu@ip-172-31-45-84:/var/www/html$ ls /var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ nano index.html
ubuntu@ip-172-31-45-84:/var/www/html$ ls
index.html
ubuntu@ip-172-31-45-84:/var/www/html$ pwd
/var/www/html
ubuntu@ip-172-31-45-84:/var/www/html$ 
```

- And refresh the browser and the Apache webpage change to your webpage



- When you're finished and ready for the Docker lab:
- sudo systemctl stop apache2
- Verify it has stopped:
- sudo systemctl status apache2

```
ubuntu@ip-172-31-45-84:/var/www/html$ sudo systemctl stop apache2
ubuntu@ip-172-31-45-84:/var/www/html$ sudo systemctl status apache2
o apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: inactive (dead) since Wed 2026-07-08 05:45:41 UTC; 51s ago
   Duration: 56min 22.616s
 Invocation: 3010c22f98ae47baae1fafbebe67eed6
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 5081 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Process: 6026 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 5081 (code=exited, status=0/SUCCESS)
     Status: "Total requests: 51; Idle/Busy workers 100/0;Requests/sec: 0.0151; Bytes served/sec:  15 B/sec"
   Mem peak: 9.4M
        CPU: 366ms

Jul 08 04:49:19 ip-172-31-45-84 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jul 08 04:49:19 ip-172-31-45-84 systemd[1]: Started apache2.service - The Apache HTTP Server.
Jul 08 05:45:41 ip-172-31-45-84 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Jul 08 05:45:41 ip-172-31-45-84 systemd[1]: apache2.service: Deactivated successfully.
Jul 08 05:45:41 ip-172-31-45-84 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-45-84:/var/www/html$
```

Build a Custom Docker Image

To Begin with the Lab

Summary of the Lab

Create a project directory, add a Dockerfile using the httpd image, and build a custom Docker image. Run the container on port 80, verify the application through the EC2 public IP, then stop the container and confirm the website becomes inaccessible.

- Now move to the directory

```
cd /home/ubuntu
```

- create a folder here
- mkdir app
- Go to the directory

```
cd app
```

- Create a file named **Dockerfile**.

```
nano Dockerfile
```

- Add the following content:

```
FROM httpd

COPY . /usr/local/apache2/htdocs
```

- Save and exit.

```
ubuntu@ip-172-31-45-84:~$ cd app
ubuntu@ip-172-31-45-84:~/app$ ls
Dockerfile  index.html
ubuntu@ip-172-31-45-84:~/app$
```

- Build the image from the current directory.

```
sudo docker build -t static-app .
```

  - docker build → Builds a Docker image.
  - -t static-app → Assigns the image name static-app.
  - . → Uses the current directory as the build context.

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker build -t static-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  5.632kB
Step 1/2 : FROM httpd
latest: Pulling from library/httpd
1eb2ba022f6d: Pulling fs layer
cfb613c6caa5: Pulling fs layer
17ff7bac6cc6: Pulling fs layer
4f4fb700ef54: Pulling fs layer
a31b3e3010ee: Pulling fs layer
e3fff84cea83: Download complete
1eb2ba022f6d: Download complete
cfb613c6caa5: Download complete
17ff7bac6cc6: Download complete
4f4fb700ef54: Download complete
cfb613c6caa5: Pull complete
4f4fb700ef54: Pull complete
a31b3e3010ee: Download complete
e7f6f34d7a39: Download complete
17ff7bac6cc6: Pull complete
1eb2ba022f6d: Pull complete
a31b3e3010ee: Pull complete
Digest: sha256:393435ee1a31437adeb1f03c134224c9ce5fa5f527e8c0cb9bea576b0d6fc742
Status: Downloaded newer image for httpd:latest
 ---> 393435ee1a31
Step 2/2 : COPY . /usr/local/apache2/htdocs
 ---> 697da8395c47
Successfully built 697da8395c47
Successfully tagged static-app:latest
ubuntu@ip-172-31-45-84:~/app$
```

- List the available Docker images.

```
sudo docker images
```

- You should see:
  - static-app, httpd (downloaded automatically if not already present) and other images present.

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker images
                                                                                  i Info →   U  In Use
IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
httpd:latest       393435ee1a31        177MB         47.6MB
nginx:latest       ec4ed8b5299e        241MB           66MB     U
static-app:latest  697da8395c47        175MB         45.2MB
ubuntu:latest      b7f48194d4d8        160MB         45.3MB     U
ubuntu@ip-172-31-45-84:~/app$
```

- List running containers.

```
sudo docker ps
```

- Stop any running containers that may already be using port **80**.

```
sudo docker stop <container-id>
```

- No container are running

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker images
                                                                                  i Info →   U  In Use
IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
httpd:latest       393435ee1a31        177MB         47.6MB
nginx:latest       ec4ed8b5299e        241MB           66MB     U
static-app:latest  697da8395c47        175MB         45.2MB
ubuntu:latest      b7f48194d4d8        160MB         45.3MB     U
ubuntu@ip-172-31-45-84:~/app$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-45-84:~/app$
```

- Create and start a container from the custom image.

```
sudo docker run -d -p 80:80 static-app
```

- Docker returns the **Container ID**.

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker run -d -p 80:80 static-app
030e114d0a9bb5c76caf8940fe76e3761b5bfc85b09a6720faad33ea966905f4
ubuntu@ip-172-31-45-84:~/app$
```

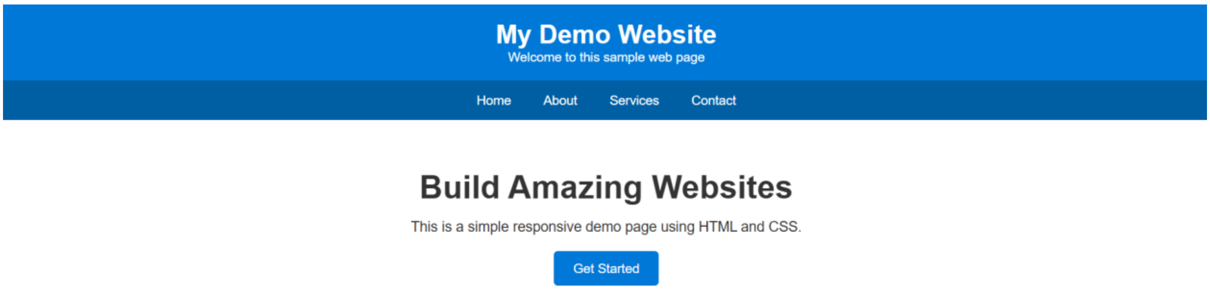
- Verify the Running Container

```
sudo docker ps
```

- Confirm that the static-app container is running.

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker run -d -p 80:80 static-app
030e114d0a9bb5c76caf8940fe76e3761b5bfc85b09a6720faad33ea966905f4
ubuntu@ip-172-31-45-84:~/app$ sudo docker ps
CONTAINER ID   IMAGE        COMMAND              CREATED          STATUS          PORTS                                 NAMES
030e114d0a9b   static-app   "httpd-foreground"   26 seconds ago   Up 26 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   nostalgic_nobel
ubuntu@ip-172-31-45-84:~/app$
```

- Open a browser and navigate to:
- http://<EC2-Public-IP>/index.html
- Your HTML web application should be displayed.



- Stop the running container.

```
sudo docker stop <container-id>
```

```
ubuntu@ip-172-31-45-84:~/app$ sudo docker stop 030
030
ubuntu@ip-172-31-45-84:~/app$
```

- Verify the Application Stops
- Refresh the browser.
- The web page will no longer load because the container has been stopped.